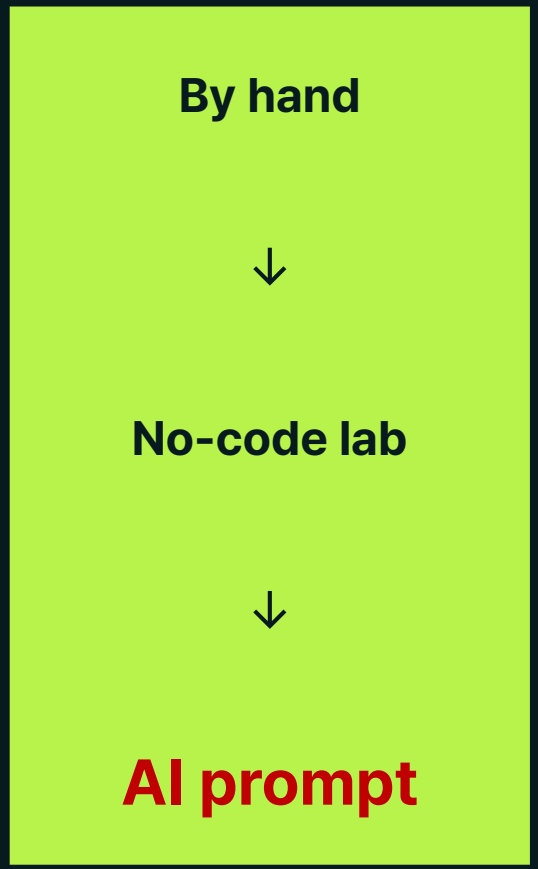
Integration, Final Project Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 16 Workbook

Four confusion-matrix counts give all four metrics

20 frauds in 1,000 — TP 8, FP 2, FN 12, TN 978

Step 1 · Accuracy · precision · recall · F1

$$\text{accuracy} = \frac{8 + 978}{1000} = 98.6\% \qquad \text{precision} = \frac{8}{8 + 2} = 80\%$$

$$\text{recall} = \frac{8}{8 + 12} = 40\% \qquad F1 = \frac{2 \times 8}{2 \times 8 + 2 + 12} = 53.3\%$$

‘All normal’ already scores 98% accuracy

Recall 40% means 60% of fraud is missed

After costs, a Sharpe of 1.2 falls to 0.72

Edge 30bp/trade, cost 12bp, 40 trades/yr, vol 10%, rf 0

Step 1 · Edge and cost

$$\text{total edge} = 30\,\text{bp} \times 40 = 12\%, \qquad \text{cost} = 12\,\text{bp} \times 40 = 4.8\%$$

Step 2 · Sharpe before/after costs

$$SR_{\text{before}} = \frac{12\%}{10\%} = 1.2, \qquad SR_{\text{after}} = \frac{7.2\%}{10\%} = 0.72$$

Step 3 · Luck max, 20 trials

$$0.5\sqrt{2\ln 20} = 0.5\sqrt{5.99} = 0.5 \times 2.448 \approx 1.22$$

Best Sharpe 1.2 is no better than the luck max 1.22

Have AI review your project report

Paste your report for a seven-item check

1. Where

Paste report into AI chat.

2. Input

A report with question, data, split, results.

3. Output

Evidence per item and a ‘cannot be verified’ list.

4. Check

Check it does not penalize simple models.

Prompt for the AI chat

You are a project reviewer for a financial ML course. Using only the report I paste, check: ① Are question, data and dates stated? ② Is there a baseline? ③ Does the split fit the data's time structure? ④ Any risk of future data, label overlap or test reuse? ⑤ Are trial count and trading costs disclosed? ⑥ Are explanations and group errors read without exaggeration? ⑦ Are reproduction steps and limits given? Cite evidence per item; mark missing ones ‘cannot be verified’. Do not deduct merely for not using a complex model.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.